



**DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING,
CEG CAMPUS, ANNA UNIVERSITY, CHENNAI – 25**



COURSE NAME: Creative and Innovative Project

COURSE CODE: CS23604

PROJECT TITLE: Video-Based Crowd Estimation Using YOLO

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1. Introduction

With the rapid expansion of urban infrastructure and public transportation systems, surveillance cameras generate vast amounts of continuous video data. Although these systems effectively record visual information, they lack automated mechanisms for interpreting crowd-related insights such as people count and crowd density. Manual monitoring of CCTV footage is inefficient, error-prone, and impractical for real-time crowd analysis.

Recent advancements in computer vision and deep learning have enabled automated understanding of complex visual scenes. Object detection models, particularly You Only Look Once (YOLO), have demonstrated high accuracy and real-time performance in detecting individuals within crowded environments. Leveraging these advancements, this project proposes a video-based crowd estimation system that automatically detects individuals, counts people, and classifies crowd density levels using CCTV footage and image datasets.

2. Problem Statement

Manual CCTV surveillance systems fail to provide accurate real-time numerical information about crowd size and density. Human operators often misjudge crowd conditions during peak hours due to occlusion, overlapping individuals, and varying camera perspectives. Traditional image-based counting methods are ineffective in dynamic video environments where scale variation and crowd movement are significant challenges.

Furthermore, many existing automated systems rely on complex retraining procedures or require extensive labeled datasets, limiting their practical applicability. Therefore, there is a need for an automated, efficient, and scalable video-based crowd estimation system capable of detecting, counting, and classifying crowd density accurately under real-world surveillance conditions.

3. Objective

The primary objectives of this project are as follows:

- To design an end-to-end computer vision pipeline for automated crowd estimation
- To perform person detection using a pre-trained YOLO deep learning model
- To extract frame-wise people counts from video sequences
- To classify crowd density levels using predefined rule-based thresholds
- To generate annotated visual outputs and quantitative performance metrics for evaluation

4. Literature Survey

Several research works have explored deep learning-based approaches for people detection and crowd monitoring.

1.Chen et al. (2025) proposed YOLO-MS, a multi-scale object detection framework using MS-Blocks and hierarchical knowledge sharing. The model improves detection accuracy for objects of varying scales and maintains real-time performance.

Limitation: The work focuses on generic object detection and does not perform people counting or crowd density interpretation.

Relation to Project: This motivates the use of YOLO for person detection while adding a separate crowd density interpretation layer.

(<https://ieeexplore.ieee.org/document/10872821>)

2.Gündüz & Işık (2023) developed a YOLO-based real-time crowd detection system using region-of-interest (ROI) analysis. Their approach enables region-wise people counting in surveillance videos.

Limitation: The system requires predefined ROI calibration, limiting adaptability across different environments.

Relation to Project: The proposed system avoids ROI dependency by performing full-frame crowd estimation.

(<https://link.springer.com/article/10.1007/s11554-023-01276-w>)

3.Vasavi et al. (2023) presented a YOLO-based people counting approach using bounding box analysis and entry–exit tracking.

Limitation: The method focuses on directional counting and does not classify crowd density levels.

Relation to Project: The proposed work extends people counting by incorporating rule-based crowd density classification.

(<https://ieeexplore.ieee.org/abstract/document/10353287>)

4.Jangale et al. (2023) proposed a YOLO-based crowd management system for CCTV surveillance.

Limitation: Computational overhead increases in dense crowd scenarios due to multiple detections and occlusions.

Relation to Project: This highlights the need for a lightweight approach, addressed in the proposed system by avoiding retraining and using simple threshold-based classification.

(https://www.researchgate.net/profile/Jay-Jangale/publication/397240794_Crowd_Management_Model_using_YOLO)

5. Proposed Work

- * Develop a video-based crowd estimation framework using deep learning
- * Extract frames from CCTV video and preprocess them for analysis
- * Detect individuals using a pre-trained YOLO person detection model

- * Perform frame-wise people counting based on detected bounding boxes
- * Classify crowd density into predefined levels using rule-based thresholds
- * Generate annotated video output with people count and density labels
- * Evaluate system performance using accuracy, robustness, and efficiency metrics.

6. New Contribution

Rule-Based Crowd Density Classification

Instead of only performing people counting, the proposed system interprets crowd conditions by classifying crowd density into four levels:

- Empty
- Moderate
- Crowded
- Jam-packed

This enables meaningful decision-making rather than providing raw numerical counts alone.

Bus-Specific Crowd Monitoring Application

The system is adapted for analyzing public transport CCTV footage, particularly bus surveillance videos.

This application-specific focus allows practical deployment in public transportation environments where crowd monitoring is critical.

Lightweight and Cost-Effective Approach

A pre-trained YOLO model is utilized without additional retraining.

This significantly reduces computational complexity and hardware requirements, making the system suitable for low-cost surveillance setups.

Video Annotation for Decision Support

The output video is annotated with:

- Bounding boxes around detected persons
- Total people count per frame
- Crowd density classification label

These annotations enhance visual interpretability and support real-time monitoring.

Proof-of-Concept Oriented System Design

The system primarily operates on recorded video inputs, enabling easier testing, debugging, and validation.

This design approach ensures the feasibility and correctness of the proposed methodology before real-time deployment.

7. Detailed Design

Video Input Module:

Accepts live CCTV feeds or pre-recorded surveillance videos and initializes video capture using OpenCV.

Frame Extraction Module:

Extracts video frames at a fixed rate and resizes them for compatibility with the detection model.

Person Detection Module:

Applies a pre-trained YOLO model to detect individuals and generate bounding boxes for the person class.

Person Counting Module:

Counts the number of detected persons in each frame based on bounding box outputs.

Crowd Density Classification Module:

Classifies crowd density into empty, moderate, crowded, or jam-packed using predefined count thresholds.

Output Visualization Module:

Displays bounding boxes, people count, and crowd density labels on the output video.

8. Evaluation Metrics

The performance of the proposed system is evaluated using the following metrics:

1. Person Detection Accuracy

- Measures the ability of the system to correctly detect individuals in video frames.
- **Metric used:**
 $\text{Detection Accuracy} = (\text{Correct Detections} / \text{Total Actual Persons}) \times 100$

2. People Counting Accuracy

- Evaluates the correctness of the estimated people count with respect to the actual count.
- **Metric used:**
 $\text{Counting Accuracy} = 1 - |\text{Actual Count} - \text{Predicted Count}| / \text{Actual Count}$

3. Crowd Density Classification Accuracy

- Measures how accurately the system classifies frames into predefined crowd density levels.
- **Metric used:**
 $\text{Classification Accuracy} = (\text{Correctly Classified Frames} / \text{Total Frames}) \times 100$

4. Robustness to Video Variations

Evaluates system performance under changes in illumination, occlusion, camera angle, and background conditions.

5. Processing Time per Frame

Measures the average time required to process a single video frame, indicating computational efficiency.

9. Test Cases

Test Case	Input Description	Expected Output
Empty Scene	No people present	Count = 0, Density = Empty
Low Crowd	6–10 persons	Accurate count, Moderate density
Medium Crowd	16–25 persons	Minor deviation, Crowded density
High Crowd	More than 30 persons	Approximate count, Jam-packed density
Occlusion	Overlapping individuals	Slight undercount, Correct density
Low Lighting	Poor illumination	Reduced confidence, Acceptable density

10. Conclusion:

This project successfully demonstrates a video-based crowd estimation system using a YOLO-based person detection model. The system accurately detects individuals, performs frame-wise people counting, and classifies crowd density levels using rule-based thresholds. Experimental results indicate reliable performance under varying crowd and video conditions with efficient processing. The proposed approach is simple, scalable, and suitable for surveillance applications. In future, the system can be extended to real-time CCTV deployment and enhanced with advanced crowd analytics.